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Tshwane University of Technology

We empower people

MOBILE COMPUTING (2025/1)

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Assignments Assignment 2

Assignment 2

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Instructions

Make sure you have reviewed all content under:

LO - 1.3 Build a basic layout using Jetpack Compose

Please create a new document containing your student number. Answer all questions provided below, making sure to indicate the question number for each response. Save the document in PDF format and upload it once completed. The PDF document should be named: <yourStudentNumber>Assign2.pdf

You may submit multiple times before the due date. Note that only the last submitted attempt will be marked.

Assignment Information

These exercises focus on how to build apps with the UI composables that you learned. The exercises a inspired by real-world use cases, some of which you probably encountered before.

These exercises provide you with resources that you need for implementation, such as images and stri The string resources contain the text that's displayed in the UI. You add these strings to the strings.xml and use them in your code.

Additionally, the exercises provide you with a set of specifications, such as font size, to use for the text content or padding around the UI components. These specifications help you build consistent UIs and often guide developers to visualize and build the screens. You might encounter similar specifications w you work with a team for an organization.

Some exercises might require you to use a Modifier. In such cases, see the **References** section availab for each problem, where you can find links to documentation related to the modifiers or properties. You can read the documentation and determine how to incorporate the concepts in the app. The ability to comprehend documentation is one of the important skills that you should develop to grow your knowledge.

You're encouraged to take as much time as you need to solve each problem thoughtfully.

Lastly, you should use Android Studio to create separate projects for these exercises.

Prerequisites

- The latest version of Android Studio
- Basic knowledge of the Kotlin programming language
- Ability to create Android projects with default templates in Android Studio.
- Knowledge of different Composable functions, such as the Text, Image, Box, Column, and Row functions
- Knowledge of Modifier classes for UI decoration

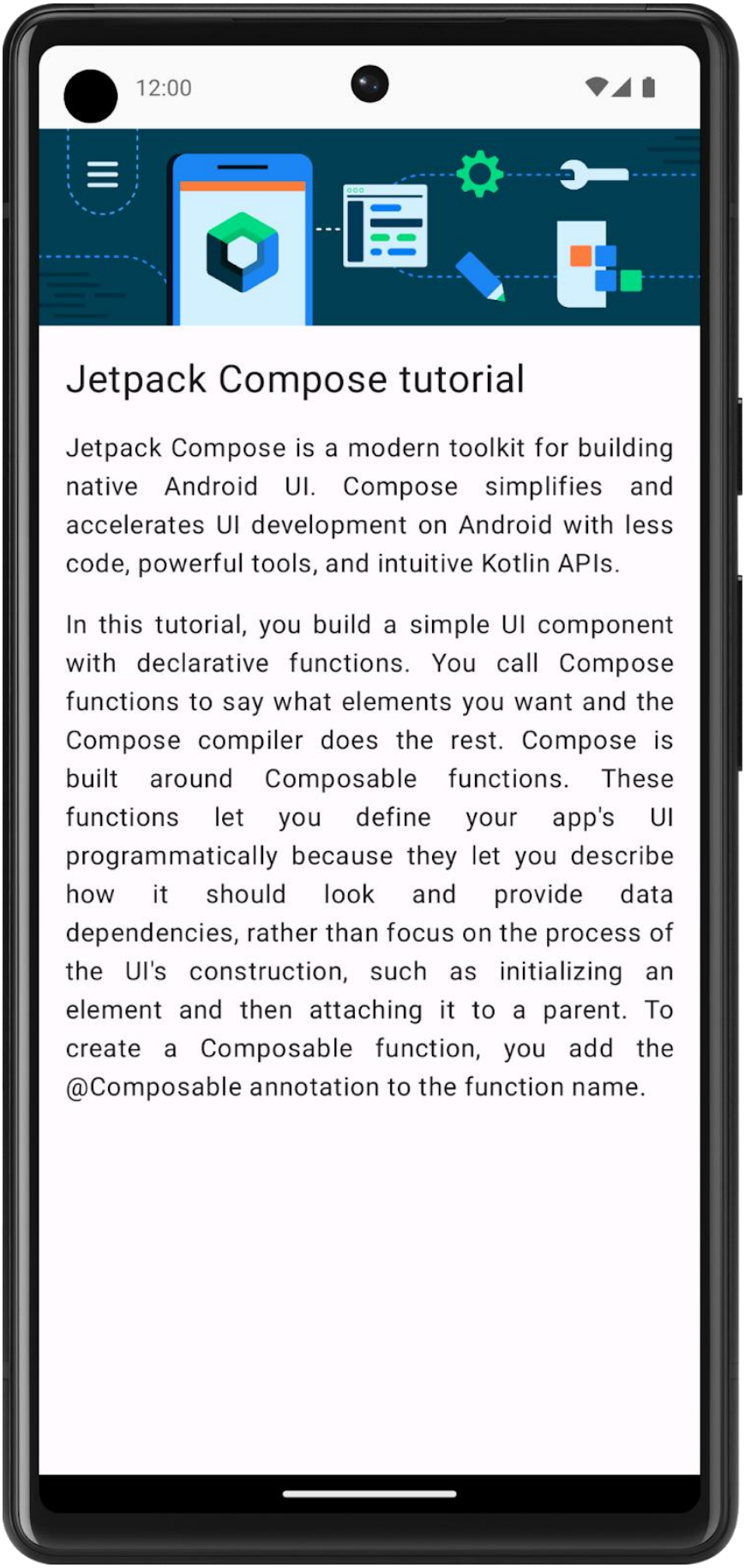
Question 1

The Learn Together app displays a list of articles about several Jetpack libraries. Users can choose the topic of their choice and learn about its latest developments.

In this question, you build a screen for the app, which displays a tutorial for Jetpack Compose. You use image and string resources provided in the **Resources** section for this problem.

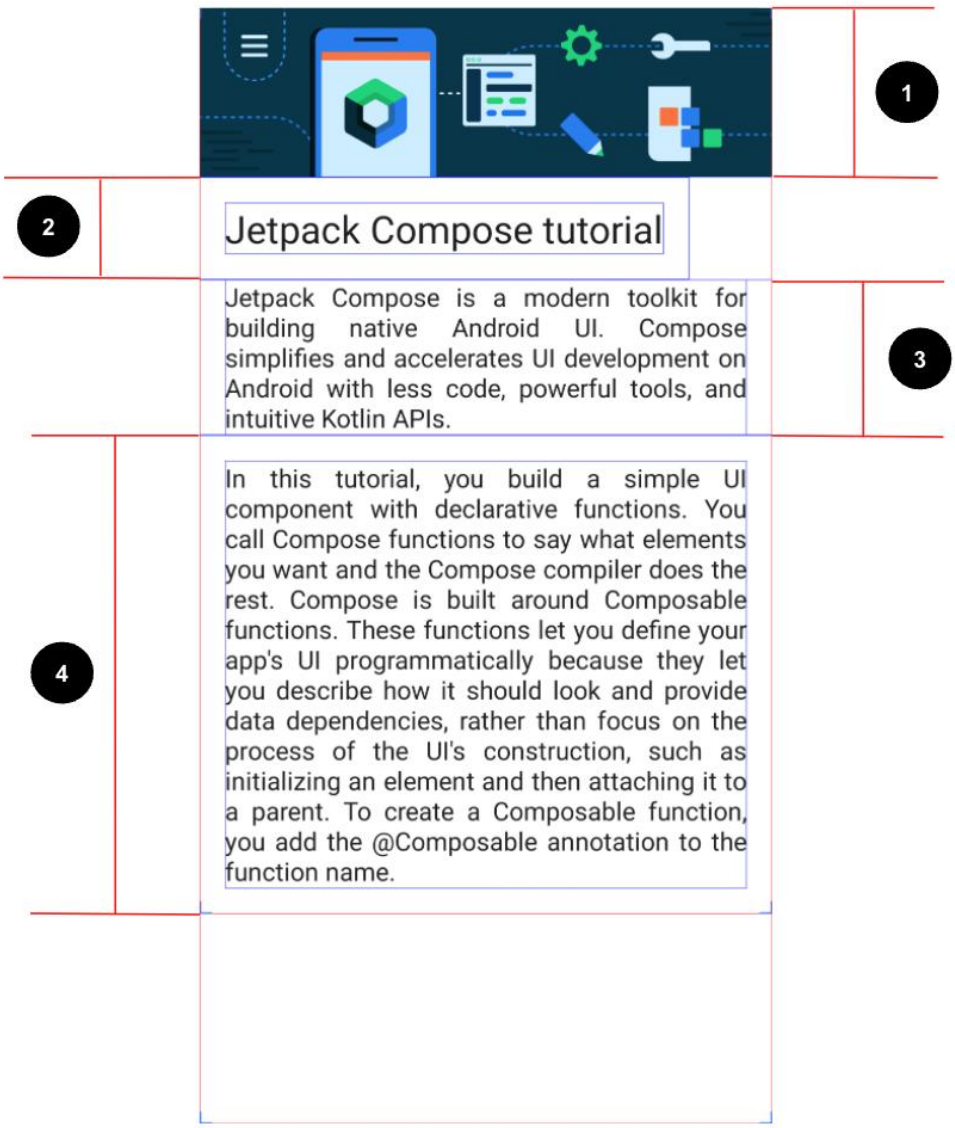
Final screenshot

After you finish the implementation, your design should match this screenshot:


12:00
Jetpack Compose tutorial
Jetpack Compose is a modern toolkit for building native Android UI. Compose simplifies and accelerates UI development on Android with less code, powerful tools, and intuitive Kotlin APIs.
In this tutorial, you build a simple UI component with declarative functions. You call Compose functions to say what elements you want and the Compose compiler does the rest. Compose is built around Composable functions. These functions let you define your app's UI programmatically because they let you describe how it should look and provide data dependencies, rather than focus on the process of the UI's construction, such as initializing an element and then attaching it to a parent. To create a Composable function, you add the @Composable annotation to the function name.

UI specifications

Follow this UI specification:



1. Set the image to fill the entire screen's width.
2. Set the first Text composable to a 24sp font size and 16dp padding (start, end, bottom, and top)
3. Set the second Text composable to a default font size, 16dp padding(start and end), and Justify align.
4. Set the third Text composable to a default font size, 16dp padding (start, end, bottom, and top), and Justify text align.

Resources

You need this image to import into your project and these strings:

- Jetpack Compose tutorial
- Jetpack Compose is a modern toolkit for building native Android UI. Compose simplifies and accelerates UI development on Android with less code, powerful tools, and intuitive Kotlin APIs
- In this tutorial, you build a simple UI component with declarative functions. You call Compose functions to say what elements you want and the Compose compiler does the rest. Compose is built around Composable functions. These functions let you define your app\'s UI programmatic because they let you describe how it should look and provide data dependencies, rather than fo on the process of the UI\'s construction, such as initializing an element and then attaching it to parent. To create a Composable function, you add the @Composable annotation to the functio name.

Hint: Which composable aligns its children vertically?

References

- TextAlign.Justify property

What to submit

Under Question 1 in your PDF document include:

1. A screenshot showing the completed interface design.
2. A screenshot showing the competed app running on your phone or emulator
3. All code your wrote as part of this question.

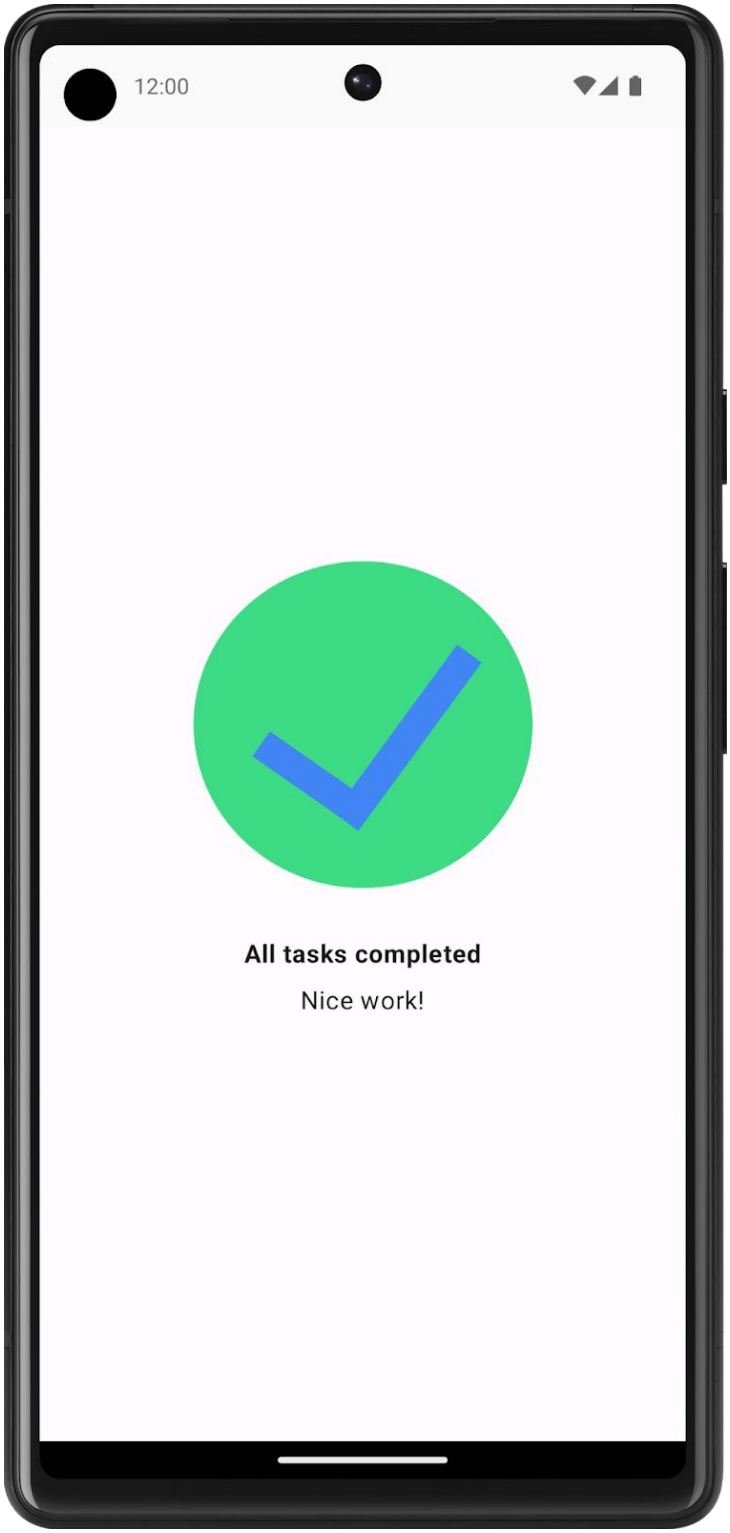
Question 2

The Task Manager app lets users manage their day-to-day tasks and check the tasks that they need to complete.

In this question, you build a screen that users see when they complete all the tasks for a given day.

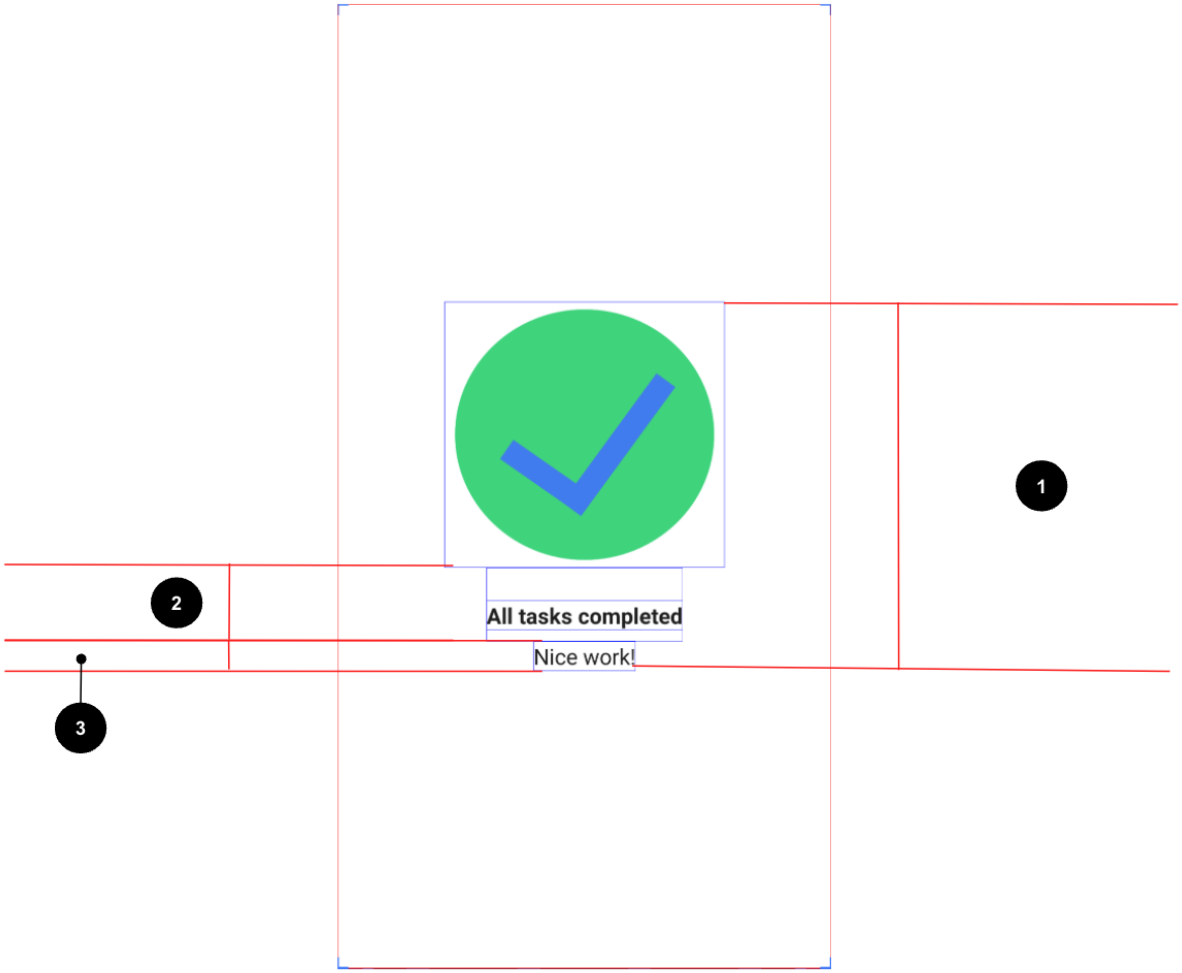
Final screenshot

After you finish the implementation, your design should match this screenshot:



UI specification

Follow these UI specifications:



1. Center align all of the content vertically and horizontally on the screen.
2. Set the first Text composable to a Bold font weight, 24dp padding top, and 8dp padding bottom.
3. Set the second Text composable to a 16sp font size.

Resources

This image, which you need to download and then import into your project, and these strings:

- All tasks completed
- Nice work!

What to submit

Under Question 2 in your PDF document include

1. A screenshot showing the completed interface design.
2. A screenshot showing the competed app running on your phone or emulator
3. All code your wrote as part of this question.

Question 3

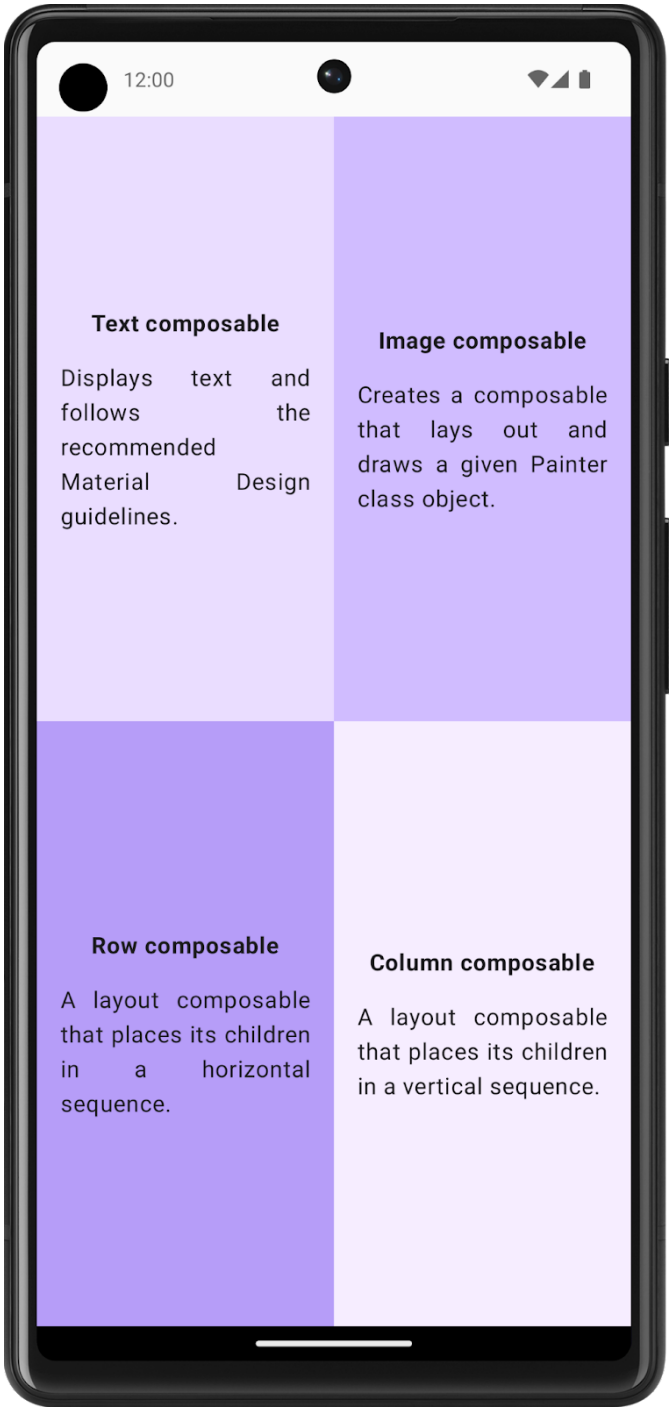
In this question, you need to apply most of the concepts that you learned so far and then go a step furt
to explore a new Modifier and properties. This might look like an extra challenge, but don't worry! You c
check the **References** section for this problem, where you can find the links to these Modifier classes
properties, and use them for implementation.

You need to build an app that displays the information about the Composable functions that you learn

The screen is divided into four quadrants. Each quadrant provides the name of a Composable function
describes it in one sentence.

Final screenshot

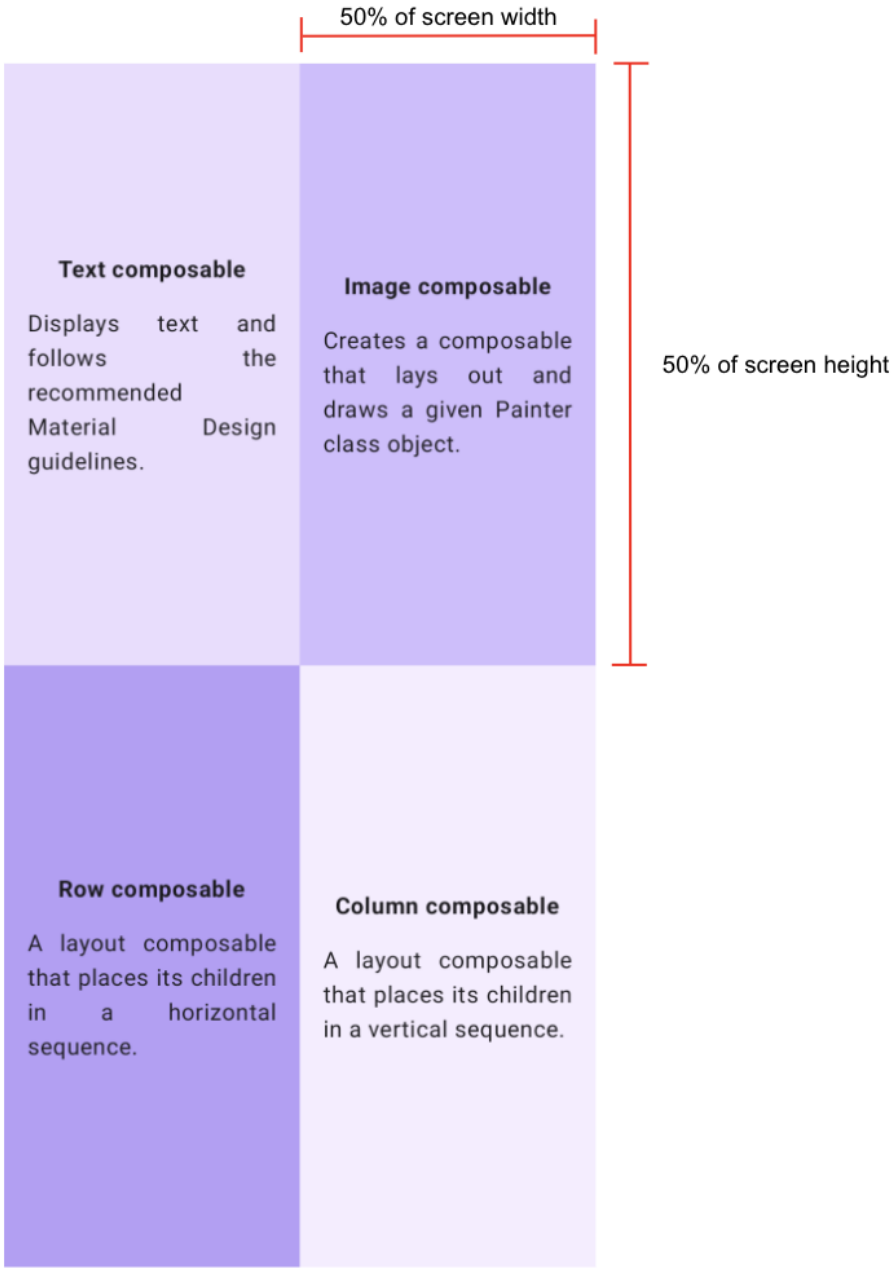
After you finish the implementation, your design should match this screenshot:



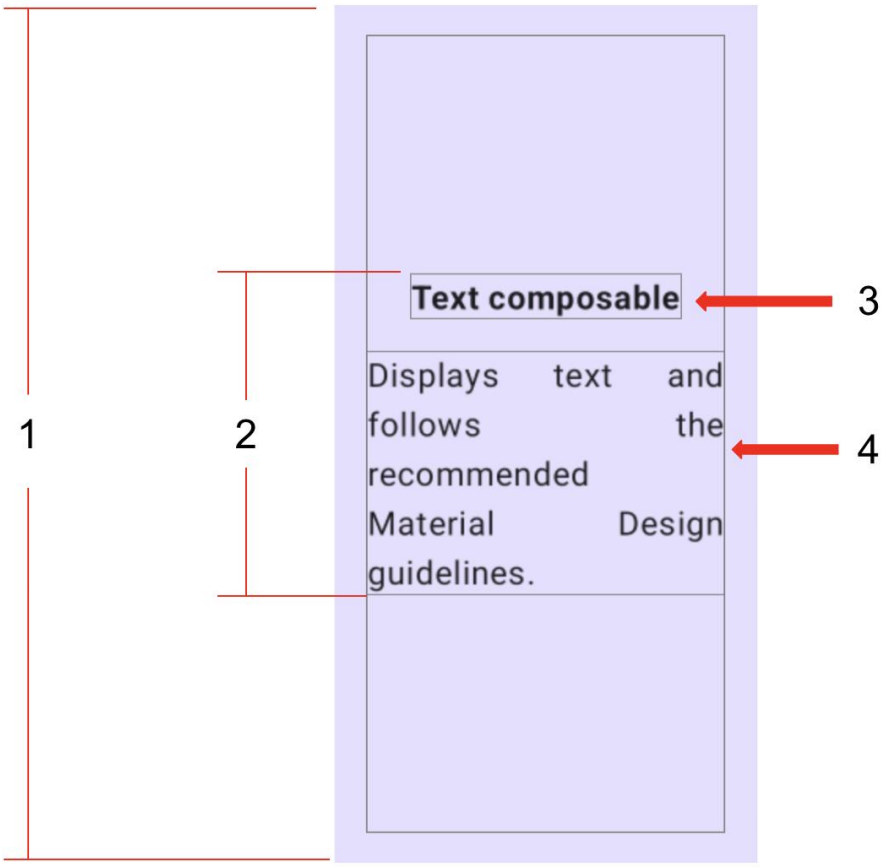
UI specifications

Follow these UI specifications for the entire screen:

- Divide the entire screen into four equal parts, each of which contains a Compose card and disp information about a Composable function.



Follow these specifications for each quadrant:



1. Set the entire quadrant (start, end, top, and bottom) to a 16dp padding.
2. Center align all of the content vertically and horizontally in each quadrant.
3. Format the first Text composable in bold and set it to a 16dp padding bottom.
4. Set the second Text composable to a Default font size.

Resources

These colors:

- Color(0xFFEADDFF)
- Color(0xFFD0BCFF)
- Color(0xFFB69DF8)
- Color(0xFFF6EDFF)

These strings:

- Text composable
- Displays text and follows the recommended Material Design guidelines.
- Image composable

- Creates a composable that lays out and draws a given Painter class object.
- Row composable
- A layout composable that places its children in a horizontal sequence.
- Column composable
- A layout composable that places its children in a vertical sequence.

References

- Weight modifier function
- FontWeight.Bold property
- TextAlign.Justify property

What to submit

Under Question 3 in your PDF document include

1. A screenshot showing the completed interface design.
2. A screenshot showing the competed app running on your phone or emulator
3. All code your wrote as part of this question.

Due on 26 February 2025 11:59 PM

Available until 26 February 2025 11:59 PM. **Access restricted after availability ends.**

Submit Assignment

Allowed File Extensions

pdf

Files to submit

(0) file(s) to submit

After uploading, you must click Submit to complete the submission.

Add a FileRecord AudioRecord Video

Comments

Paragraph

Lato (Recommended)

19px (Default)

SubmitCancel

jiijdfvkjfdvhjhdfuidsuidsidsusosmrrrrrrrr